

## 1 • INCLUSION BODIES — THE MASTER TABLE ★★★

| Named body             | Site                                | Disease                                                                 |
|------------------------|-------------------------------------|-------------------------------------------------------------------------|
| Negri body             | Intra-CYTOPLASMIC                   | <b>RABIES</b> — hippocampal pyramidal cells & cerebellar Purkinje cells |
| Cowdry type A          | Intra-NUCLEAR                       | <b>Herpesviruses</b> (IBR, EHV-1) and <b>adenovirus (ICH)</b>           |
| Bollinger body         | Intra-cytoplasmic                   | <b>FOWL POX</b> (contains Borrel bodies = elementary bodies)            |
| Guarnieri body         | Intra-cytoplasmic                   | Vaccinia / cowpox / buffalopox                                          |
| Borrel body            | Intra-cytoplasmic                   | Elementary bodies of pox viruses                                        |
| CDV inclusions         | <b>BOTH nuclear AND cytoplasmic</b> | <b>Canine distemper</b> — epithelium, astrocytes, neurons, bladder      |
| Marek's — no inclusion | —                                   | Lymphoproliferative, not inclusion-forming                              |

**RULE OF THUMB** DNA viruses replicate in the nucleus → **INTRANUCLEAR** inclusions (herpes, adeno, parvo). **Exception: POXVIRUS** is a DNA virus that replicates in the **CYTOPLASM** → **intracytoplasmic**. RNA viruses → **intracytoplasmic** (rabies, distemper) — but paramyxoviruses (CDV, rinderpest) can give **both**.

## 2 • FOOT AND MOUTH DISEASE ★★★

- **Aphthovirus**, family **Picornaviridae**; ssRNA. **7 serotypes** — **O, A, C, SAT-1, SAT-2, SAT-3, Asia-1**. In India the circulating types are **O, A and Asia-1**. **No cross-protection between serotypes** — this is why vaccination fails when the type changes
- Cloven-footed animals. **Very high morbidity (up to 100%), low mortality in adults** — but **high mortality in young stock**
- **Gross: vesicles (aphthae)** → ruptured, leaving red raw erosions, on the **tongue, dental pad, gums, muzzle, interdigital space, coronary band and teats**; profuse ropy salivation, lip-smacking, lameness
- **"TIGER HEART"** ★ in calves — pale yellow-grey streaks of myocardial degeneration and necrosis alternating with red muscle; this is what kills the young
- **Micro: intra-epithelial vesicle** from **hydropic and ballooning degeneration of the stratum spinosum**
- **NOTIFIABLE**. Sample: **vesicular fluid and epithelial flap in 50% glycerol-phosphate buffer, pH 7.2–7.6**, sent chilled

**THE 4 VESICULAR DISEASES LOOK IDENTICAL ★★ FMD** — cattle, sheep, goat, pig; **NOT horses**. **Vesicular stomatitis** — the only one that **DOES affect HORSES** (and is zoonotic). **Vesicular exanthema** and **swine vesicular disease** — **pigs only**. They can be separated only in the laboratory. **The horse is the field clue**.

## 3 • RINDERPEST • PPR • MCF

| Disease                           | Agent                                                                                                                      | Signature lesion                                                                                                                                                                                                                                                                                                                                              |
|-----------------------------------|----------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>RINDERPEST</b> (cattle plague) | <b>Morbillivirus</b> , Paramyxoviridae                                                                                     | <b>"ZEBRA STRIPING"</b> — linear congestion and haemorrhage on the longitudinal folds of the <b>colon and rectum</b> . Necrotic erosive stomatitis, "shooting" diarrhoea, <b>Peyer's patch necrosis</b> , syncytial giant cells with <b>both nuclear and cytoplasmic inclusions</b> . <b>Globally eradicated (declared 2011); India declared free in 2006</b> |
| <b>PPR</b> (goat plague)          | <b>Morbillivirus</b>                                                                                                       | <b>Stomatitis–pneumoenteritis of goats more than sheep</b> . Erosive stomatitis, mucopurulent oculo-nasal discharge, <b>zebra striping of colon, bronchointerstitial pneumonia</b> . Looks like rinderpest but in small ruminants                                                                                                                             |
| <b>MALIGNANT CATARRHAL FEVER</b>  | <b>Ovine herpesvirus-2</b> (sheep-associated, the Indian form) or <b>alcelaphine herpesvirus-1</b> (wildebeest-associated) | <b>BILATERAL CORNEAL OPACITY spreading from the LIMBUS INWARD</b> — the classic "head and eye" form. Generalised <b>lymphadenopathy</b> , erosive stomatitis and rhinitis. <b>THE microscopic key: widespread lymphocytic NECROTISING VASCULITIS/ARTERITIS in every organ</b> . Sporadic, <b>almost always fatal</b> , cattle are dead-end hosts              |

## 4 • BLUETONGUE • IBR • BVD-MD

| Disease                        | Agent & key facts                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|--------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>BLUETONGUE</b>              | <b>Orbivirus</b> , <b>Reoviridae</b> ; transmitted by <b>Culicoides</b> midges, therefore <b>seasonal (rainy season) and non-contagious</b> . <b>SHEEP</b> are worst affected. Facial and submandibular <b>oedema, cyanotic swollen protruding tongue</b> , erosions of the buccal mucosa, <b>coronitis with a red-purple band at the coronary band</b> and later hoof separation, torticollis. ★ <b>Haemorrhage at the BASE OF THE PULMONARY ARTERY is the characteristic post-mortem finding</b> . Skeletal muscle necrosis                                                                                                                                                                                                                                                                             |
| <b>IBR / IPV</b>               | <b>Bovine herpesvirus-1</b> . <b>Respiratory form</b> ("red nose") — necrotic grey-white plaques on the nasal mucosa, laryngotracheitis with a fibrinonecrotic cast. <b>Genital form</b> — infectious pustular vulvovaginitis / balanoposthitis. <b>Abortion</b> — usually in the last trimester, <b>autolysed fetus with pinpoint necrotic foci in the liver</b> . Also conjunctivitis and calf encephalitis. <b>Cowdry type A intranuclear inclusions</b> early. Establishes <b>LATENCY in the trigeminal ganglion</b> and recrudesces with stress                                                                                                                                                                                                                                                      |
| <b>BVD / MUCOSAL DISEASE ★</b> | <b>Pestivirus</b> , <b>Flaviviridae</b> . Two biotypes: <b>non-cytopathic (ncp)</b> and <b>cytopathic (cp)</b> . <b>The PI story — learn this:</b> infection of the dam with the <b>ncp</b> biotype at about <b>40–120 days of gestation</b> , before the fetal immune system matures, produces a calf that is <b>IMMUNOTOLERANT and PERSISTENTLY INFECTED (PI)</b> . That PI animal sheds virus for life and develops <b>fatal MUCOSAL DISEASE if it is later superinfected by the CYTOPATHIC biotype</b> . <b>Lesions:</b> linear erosions/ulcers of the oesophagus (highly suggestive), erosions from mouth to colon, <b>Peyer's patch necrosis</b> , interdigital erosions, lymphoid depletion. Infection at about <b>100–170 days</b> instead causes <b>CEREBELLAR HYPOPLASIA</b> and ocular defects |

## 5 • EQUINE VIRAL DISEASES

| Disease                                          | Agent & lesion                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|--------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Equine infectious anaemia</b> ("swamp fever") | <b>Lentivirus, Retroviridae</b> ; biting flies and contaminated needles. <b>Recurrent fever, progressive anaemia, ventral oedema, emaciation; LIFELONG CARRIER.</b> Pathogenesis is <b>immune-mediated</b> : repeated antigenic drift → immune complexes → <b>Type III glomerulonephritis</b> and Coombs-positive haemolysis. Micro: <b>SIDEROLEUKOCYTES</b> (haemosiderin-laden macrophages) in liver, spleen and bone marrow; lymphoid proliferation around vessels. <b>Diagnosis: Coggins test (AGID)</b> and cELISA |
| <b>Equine influenza</b>                          | <b>Influenza A</b> , subtypes <b>H7N7 and H3N8</b> . Sudden <b>dry, harsh, hacking cough</b> , high fever, rapid spread; damage to the tracheal cilia predisposes to secondary bacterial pneumonia                                                                                                                                                                                                                                                                                                                      |
| <b>Equine viral arteritis</b>                    | <b>Arterivirus.</b> Panvascular disease — <b>necrotising arteritis of small muscular arteries</b> → oedema of limbs, ventral abdomen and periorbital tissue ("pink eye"), <b>abortion</b> at any stage. <b>Stallions become long-term carriers and shed in semen</b>                                                                                                                                                                                                                                                    |
| <b>Equine rhinopneumonitis</b>                   | <b>EHV-1 and EHV-4. EHV-4</b> → mainly upper respiratory disease in foals. <b>EHV-1</b> → <b>abortion storms in the last trimester</b> ; the aborted fetus is <b>FRESH (not autolysed)</b> , icteric, with pleural fluid and <b>pinpoint white necrotic foci in the liver with Cowdry type A intranuclear inclusions</b> ; also <b>myeloencephalopathy</b> from vasculitis and thrombosis                                                                                                                               |
| <b>Equine encephalomyelitis</b>                  | <b>Alphavirus (Togaviridae)</b> — <b>EEE, WEE, VEE</b> ; mosquito-borne, birds are the reservoir, <b>zoonotic. Non-suppurative polioencephalomyelitis</b> — neuronal necrosis, neuronophagia, perivascular cuffing, gliosis, chiefly in <b>grey matter</b>                                                                                                                                                                                                                                                              |

## 6 • SWINE VIRAL DISEASES

**CLASSICAL SWINE FEVER — THE FOUR-LESION SET ★★★** Pestivirus (same genus as BVD). Highly contagious, high mortality.

- ① **"BUTTON ULCERS" (botryomycotic buttons)** — raised, laminated, concentric necrotic ulcers of the **caecum, colon and ileocaecal valve**
- ② **MARGINAL SPLENIC INFARCTS** — dark wedge-shaped infarcts on the splenic border (near pathognomonic)
- ③ **"TURKEY-EGG KIDNEY"** — petechiae scattered over a pale renal cortex; petechiae also in bladder, larynx and skin
- ④ **HAEMORRHAGIC ("MARBLED") LYMPH NODES** — haemorrhage at the periphery of the node  
Plus **non-suppurative encephalitis with perivascular cuffing**.  
Congenital infection → **cerebellar hypoplasia and congenital tremor**

**Swine influenza** — Influenza A, usually **H1N1**; explosive outbreak, high morbidity, low mortality; **cranioventral purple-red consolidation**, necrotising bronchiolitis; **zoonotic potential**.

## 7 • CANINE &amp; FELINE VIRAL DISEASES ★★

| Disease                            | Agent & key lesion                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>RABIES</b>                      | <b>Lyssavirus, Rhabdoviridae, bullet-shaped.</b> Virus in <b>saliva</b> → <b>centripetal</b> spread along <b>peripheral nerves</b> to the CNS, then <b>centrifugal</b> spread to the <b>salivary glands</b> (that is why the bite is infective). Forms: <b>furious, dumb/paralytic. Gross lesions are minimal or absent</b> — foreign bodies in a dog's stomach are suggestive. Micro: <b>non-suppurative polioencephalitis + NEGRI BODIES</b> (intracytoplasmic, in <b>hippocampal pyramidal neurons</b> of carnivores and <b>cerebellar Purkinje cells</b> of herbivores); <b>Babes' nodules</b> = glial nodules.<br><b>Diagnosis: FAT on brain impression smear is the GOLD STANDARD</b> ; also <b>Seller's stain</b> for Negri bodies, mouse inoculation, RT-PCR. Zoonosis — 100% fatal once signs appear |
| <b>CANINE DISTEMPER</b>            | <b>Morbillivirus.</b> Pantropic — respiratory, GI, skin, CNS. <b>Biphasic fever</b> , oculo-nasal discharge, <b>bronchointerstitial pneumonia, "HARD PAD" (nasodigital hyperkeratosis), enamel hypoplasia ("distemper teeth")</b> if infected before teeth erupt, <b>demyelinating leukoencephalitis</b> , "chewing-gum" fits. <b>Inclusions are BOTH intranuclear and intracytoplasmic</b> , best seen in <b>bladder, gastric and bronchial epithelium and astrocytes</b> . Marked <b>lymphoid depletion</b> → <b>secondary infection</b>                                                                                                                                                                                                                                                                    |
| <b>INFECTIOUS CANINE HEPATITIS</b> | <b>Canine adenovirus-1. Centrilobular hepatic necrosis</b> with <b>Cowdry type A intranuclear inclusions</b> in hepatocytes and <b>endothelium</b> ; <b>OEDEMA OF THE GALL-BLADDER WALL is characteristic</b> ; fibrin strands on the liver capsule; petechiae; <b>"BLUE EYE"</b> = corneal oedema and anterior uveitis, a <b>Type III immune-complex</b> reaction during recovery or after live vaccine                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>CANINE PARVOVIRUS (CPV-2)</b>   | Attacks <b>rapidly dividing cells. Intestinal CRYPT NECROSIS</b> (not the villus tip) → villus collapse, <b>haemorrhagic enteritis</b> with foul bloody diarrhoea and vomiting; <b>lymphoid depletion and marked leucopenia</b> ; <b>myocarditis</b> in pups infected under about 8 weeks. Intranuclear inclusions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>FELINE PANLEUKOPENIA</b>        | Feline parvovirus. <b>Severe panleukopenia</b> , crypt necrosis and enteritis, and <b>CEREBELLAR HYPOPLASIA</b> in kittens infected in utero or just after birth                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |

## 8 • OVINE RETROVIRAL LUNG DISEASE • ROTA &amp; CORONA • PRIONS

| Disease                 | Key point                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>MAEDI (– visna)</b>  | <b>Ovine lentivirus. Maedi</b> = chronic <b>progressive INTERSTITIAL pneumonia</b> — lungs heavy, grey-blue, rubbery, <b>DO NOT COLLAPSE</b> , with rib impressions and lymphoid follicle formation. <b>Visna</b> = <b>demyelinating meningoencephalitis</b> . Long incubation, <b>no fever, no cure</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>JAAGSIEKTE ★</b>     | <b>Ovine pulmonary adenocarcinoma</b> , a <b>betaretrovirus</b> — a genuinely <b>CONTAGIOUS LUNG TUMOUR</b> . Firm grey-white nodules, later confluent; abundant fluid. <b>"WHEELBARROW TEST"</b> — lifting the hindquarters makes <b>copious frothy fluid pour from the nostrils</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Rotavirus</b>        | Neonatal diarrhoea. Infects <b>MATURE enterocytes at the VILLUS TIP</b> → <b>villous atrophy and blunting</b> → <b>malabsorptive, osmotic diarrhoea</b> . Crypts are spared, so recovery is possible                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Coronavirus</b>      | <b>TGE of pigs</b> — very severe villous atrophy, near-100% mortality under 2 weeks. <b>Bovine coronavirus</b> — calf diarrhoea and <b>winter dysentery</b> in adults. Also feline coronavirus → <b>FIP</b> with pyogranulomatous vasculitis                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>PRION DISEASES ★</b> | The agent is <b>PrP<sup>Sc</sup></b> — an <b>abnormally folded PROTEIN with NO nucleic acid</b> . Therefore: <b>no inflammation, no immune response, no antibody, no fever</b> , and extreme resistance to <b>formalin, heat, UV and ordinary disinfectants</b> . Very long incubation, invariably fatal.<br><b>Common lesion: STATUS SPONGIOSUS</b> — vacuolation of neurons and neuropil in the grey matter, gliosis, neuronal loss, <b>amyloid plaques</b> .<br><b>SCRAPIE</b> (sheep, goats) — intense <b>pruritus</b> , wool loss, ataxia. <b>BSE</b> (cattle, "mad cow") — spread by feeding <b>meat-and-bone meal</b> ; apprehension, hyperaesthesia, ataxia; linked to variant CJD in people. <b>FSE</b> in cats; <b>chronic wasting disease</b> in deer.<br><b>Diagnosis: histopathology of the brain stem (obex) + IMMUNOHISTOCHEMISTRY or Western blot for PrP</b> |

## 9 • POX VIRUS DISEASES

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |        |   |         |   |             |   |                 |   |      |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|--------|---|---------|---|-------------|---|-----------------|---|------|
| Macul<br>e                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | → | Papule | → | Vesicle | → | Pustul<br>e | → | Crust /<br>scab | → | Scar |
| <ul style="list-style-type: none"> <li>• <b>Poxviruses are large DNA viruses that replicate in the CYTOPLASM</b>, hence <b>intracytoplasmic inclusions</b>; micro shows <b>ballooning degeneration</b> of keratinocytes and epidermal hyperplasia</li> <li>• <b>SHEEP POX</b> — the <b>most severe</b> pox of domestic animals (Capripoxvirus); generalised skin nodules, and <b>hard grey nodules in the LUNG</b> are characteristic; "sheep-pox cells" with vacuolated nuclei</li> <li>• <b>GOAT POX</b> — similar, milder; <b>LUMPY SKIN DISEASE</b> (capripox, cattle) — multiple firm intradermal nodules over the whole body, "sit-fast" cores, lymphadenopathy, oedema of the limbs</li> <li>• <b>Cowpox / BUFFALOPOX</b> — teat and udder lesions, <b>zoonotic</b>, important in Indian dairies (milkers' nodules)</li> <li>• <b>ORF (contagious ecthyma / contagious pustular dermatitis)</b> — a <b>PARAPOXVIRUS</b> of sheep and goats; <b>proliferative, scabby, wart-like lesions on the lips, commissures and muzzle</b> of lambs; <b>zoonotic</b>; interferes with suckling</li> <li>• <b>Swine pox</b> — usually spread by the <b>pig louse <i>Haematopinus suis</i></b>;</li> <li>• <b>Fowl pox</b> — cutaneous (comb, wattle) and diphtheritic forms, <b>Bollinger bodies</b></li> </ul> |   |        |   |         |   |             |   |                 |   |      |

## 10 • TUBERCULOSIS &amp; JOHNE'S DISEASE ★★★

|              | <b>Tuberculosis</b>                                                                                                                                                      | <b>Johne's disease</b>                                                                                                                                                         |
|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Agent        | <b><i>M. bovis</i></b> (cattle), <b><i>M. tuberculosis</i></b> , <b><i>M. avium</i></b>                                                                                  | <b><i>M. avium</i> subsp. <i>paratuberculosis</i></b>                                                                                                                          |
| Reaction     | <b>Granuloma (tubercle) with CASEOUS necrosis + epithelioid cells + Langhans giant cells + lymphocyte cuff + fibrous capsule; calcification is prominent in cattle</b>   | <b>DIFFUSE granulomatous enteritis — NO caseation, NO calcification, NO ulceration; sheets of epithelioid macrophages stuffed with acid-fast bacilli in the lamina propria</b> |
| Gross        | <b>Miliary nodules; "PEARL DISEASE"</b> = grape-like nodules on pleura and peritoneum; <b>Ghon (primary) complex</b> = the primary lung lesion + its regional lymph node | <b>Thickened, CORRUGATED / rugose ileum that will not flatten, "CORDING" of the subserosal lymphatics, enlarged oedematous mesenteric lymph nodes</b>                          |
| Signs        | Chronic wasting, cough                                                                                                                                                   | <b>Chronic intractable diarrhoea with a normal appetite, progressive emaciation, bottle jaw; adults, long incubation</b>                                                       |
| Stain / test | <b>Ziehl–Neelsen ACID-FAST; tuberculin test = Type IV delayed hypersensitivity</b>                                                                                       | <b>ZN acid-fast in clumps; johnin test, faecal culture, ELISA, PCR</b>                                                                                                         |

## 11 · ANTHRAX &amp; THE CLOSTRIDIA ★★★

**ANTHRAX** *Bacillus anthracis* — Gram-positive, **capsulated, spore-forming** rod; spores survive in soil for decades. **Peracute death** in ruminants. **Dark tarry blood that fails to clot oozes from the nostrils, mouth, anus and vulva; rigor mortis is absent or incomplete; the SPLEEN is huge, soft and blackberry-jam-like.** Widespread haemorrhage and oedema.

**NEVER open the carcass.** Make a **peripheral blood smear** and stain with **polychrome methylene blue** → **McFADYEAN'S REACTION** (blue rods with a pink capsule). Dispose by **deep burial with quicklime or incineration**. Notifiable, zoonotic (malignant pustule, woolsorter's disease). In **pigs** it is usually a localised **pharyngeal/oedematous** form; in horses, colic and oedema.

| Clostridial disease                                  | Organism & the fact that identifies it                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>BLACK QUARTER (blackleg)</b>                      | <b><i>Cl. chauvoei</i></b> . <b>ENDOGENOUS</b> — spores already in the muscle germinate after bruising, <b>no wound needed</b> . <b>Young, well-nourished cattle 6 months–2 years</b> . Heavy muscle becomes <b>dark red-black, dry, spongy, CREPITANT</b> with a <b>rancid-butter odour</b> ; hot painful swelling that becomes cold and crackly. Prevented by vaccination before the rains                                                                                            |
| <b>Malignant oedema</b>                              | <b><i>Cl. septicum</i></b> — <b>EXOGENOUS, needs a WOUND</b> ; more <b>oedematous, less gassy</b> than blackleg                                                                                                                                                                                                                                                                                                                                                                         |
| <b>BLACK DISEASE</b> (infectious necrotic hepatitis) | <b><i>Cl. novyi</i> type B</b> in sheep. <b>Requires liver damage by migrating immature <i>Fasciola</i> to create the anaerobic focus. Large pale-yellow necrotic foci surrounded by a hyperaemic zone in the liver; after death intense subcutaneous venous congestion turns the skin dark</b> — hence the name                                                                                                                                                                        |
| <b>ENTEROTOXAEMIA</b>                                | <b><i>Cl. perfringens</i> types. Type D = "PULPY KIDNEY" of lambs</b> — <b>epsilon toxin</b> after a sudden change to rich feed; <b>GLYCOSURIA</b> is a useful antemortem clue; kidneys soften rapidly after death; <b>focal symmetrical encephalomalacia</b> . <b>Type B = lamb dysentery; Type C = "struck"</b> and necrotic enteritis                                                                                                                                                |
| <b>BRAXY</b>                                         | <b><i>Cl. septicum</i></b> — <b>ABOMASITIS</b> of sheep that eat <b>frosted/frozen grass</b> ; haemorrhagic necrotic abomasal wall                                                                                                                                                                                                                                                                                                                                                      |
| <b>BOTULISM</b>                                      | <b><i>Cl. botulinum</i></b> — an <b>INTOXICATION</b> : the <b>preformed toxin is EATEN</b> . Toxin <b>blocks acetylcholine release</b> at the neuromuscular junction → <b>FLACCID paralysis</b> , tongue paralysis, "limberneck" in birds. <b>Type D in cattle follows PHOSPHORUS deficiency</b> → <b>osteophagia</b> (chewing carcass bones)                                                                                                                                           |
| <b>TETANUS</b>                                       | <b><i>Cl. tetani</i></b> — enters through a <b>deep puncture wound</b> . <b>Tetanospasmin blocks the release of the inhibitory transmitters glycine and GABA</b> from the Renshaw cells → <b>SPASTIC paralysis</b> : lockjaw (trismus), " <b>sawhorse</b> " stance, erect ears, <b>protrusion of the third eyelid</b> , spasms triggered by noise or touch; <b>the animal remains conscious</b> . <b>Horses are the most susceptible</b> ; there are <b>no diagnostic gross lesions</b> |

**TWO TOXINS, OPPOSITE RESULTS** **BOTULISM = FLACCID** (ACh release blocked). **TETANUS = SPASTIC/RIGID** (inhibition blocked). Both are ***Clostridium*** neurotoxins.

## 12 · ACTINOMYCOSIS · STREPTOCOCCI · STAPHYLOCOCCI · GLANDERS

- **ACTINOMYCOSIS** — "lumpy jaw": ***Actinomyces bovis***, a rarefying and proliferative **osteomyelitis of the MANDIBLE**; pyogranuloma containing "**sulphur granules**" (club colonies) with the **Splendore–Hoeppli** reaction
- **ACTINOBACILLOSIS** — "wooden tongue": ***Actinobacillus lignieresii***, in **SOFT tissue** — the tongue becomes hard and protrudes; same club colonies
- **STRANGLES**: ***Streptococcus equi* subsp. *equi*** in young horses — fever, nasal discharge and **abscessation of the submandibular and retropharyngeal lymph nodes** which rupture and discharge thick pus. "**Bastard strangles**" = metastatic abscesses in other organs. **PURPURA HAEMORRHAGICA** is the **Type III immune-complex sequel** — widespread petechiae and severe oedema
- **STAPHYLOCOCCOSIS**: ***Staph. aureus*** — abscess, **gangrenous mastitis**, tick pyaemia of lambs, **botryomycosis** (chronic granuloma with club colonies)
- **GLANDERS** ★: ***Burkholderia mallei***, of equines, **zoonotic and notifiable in India**. **Nasal form** — ulcers on the septum that heal into a typical **stellate (star-shaped) scar**. **Pulmonary form** — miliary grey nodules. **Cutaneous form = "FARCY"** — **corded, thickened lymphatics studded with nodules ("farcy buds") that ulcerate**. Diagnosed by the **MALLEIN test** and complement fixation

## 13 · PASTEURELLOSIS ★★★ (TOP INDIAN TOPIC)

|           | Haemorrhagic septicaemia                                                                                                                                                                                                                                                          | Shipping fever / pneumonic pasteurellosis                                                                                                                        |
|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Agent     | <b><i>Pasteurella multocida</i> serotypes B:2 and E:2</b>                                                                                                                                                                                                                         | <b><i>Mannheimia (Pasteurella) haemolytica</i> A1</b>                                                                                                            |
| Species   | <b>Cattle and especially BUFFALO</b> ; monsoon disease of India                                                                                                                                                                                                                   | Cattle, sheep after transport/stress                                                                                                                             |
| Course    | <b>Peracute septicaemia</b> , death in 6–24 h                                                                                                                                                                                                                                     | Acute bronchopneumonia                                                                                                                                           |
| Lesions   | <b>Hot painful SUBCUTANEOUS OEDEMA of the throat, brisket and dewlap</b> (the "septicaemic/oedematous" form); widespread <b>petechiae on serosae</b> ; blood-tinged fluid in the body cavities; enlarged haemorrhagic lymph nodes; <b>fibrinous pneumonia</b> in the longer cases | <b>CRANIOVENTRAL fibrinous bronchopneumonia with FIBRINOUS PLEURITIS</b> ; interlobular oedema; the lung is red, firm and heavy with fibrin sheets on the pleura |
| Diagnosis | <b>Bipolar organisms on a Leishman-stained blood smear</b> ; culture from heart blood, spleen, oedema fluid                                                                                                                                                                       | Culture from lung; history of transport                                                                                                                          |

**FIELD RULE** A buffalo that dies in **24 hours during the rains with a swollen throat = HS**. A buffalo found dead with **tarry unclotted blood and a huge soft spleen = ANTHRAX**. Do not confuse them — the second one you must not open.

## 14 · LEPTOSPIROSIS · LISTERIOSIS · SWINE ERYSIPELAS · BRUCELLOSIS

| Disease                 | Agent & lesions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>LEPTOSPIROSIS</b>    | <i>Leptospira interrogans</i> serovars — <i>pomona</i> , <i>hardjo</i> , <i>canicola</i> , <i>icterohaemorrhagiae</i> , <i>grippotyphosa</i> . Spread in <b>URINE</b> , enters through wet skin and mucosa; <b>rodents are the reservoir</b> . Acute in calves — <b>intravascular haemolysis with haemoglobinuria ("REDWATER")</b> , jaundice and fever. Chronic — <b>interstitial nephritis giving the "WHITE-SPOTTED KIDNEY"</b> , abortion, and <b>agalactia with flabby udder in cows</b> . In dogs, acute nephritis and hepatitis. Diagnosis: <b>dark-field microscopy of urine</b> , <b>MAT</b> , silver stain of kidney, PCR. <b>Important ZOONOSIS</b> (Weil's disease)                                                                                                                                                                                                                     |
| <b>LISTERIOSIS</b>      | <i>Listeria monocytogenes</i> — Gram-positive, grows at refrigeration temperature ( <b>cold enrichment</b> ) and multiplies in <b>poorly fermented SILAGE with pH above about 5.5</b> . Three syndromes: ① <b>"CIRCLING DISEASE"</b> — ascends the <b>trigeminal nerve</b> to give <b>MICROABSCESSES IN THE BRAIN STEM (pons and medulla)</b> with <b>unilateral</b> cranial nerve deficits, drooping ear and lip, circling. ② <b>Abortion</b> in the last trimester with focal hepatic necrosis in the fetus. ③ <b>Septicaemia</b> with miliary hepatic necrosis in monogastrics and neonates. <b>Zoonotic</b>                                                                                                                                                                                                                                                                                     |
| <b>SWINE ERYSIPELAS</b> | <i>Erysipelothrix rhusiopathiae</i> . <b>Acute</b> — septicaemia with <b>DIAMOND-SHAPED (rhomboid) raised red-purple SKIN LESIONS</b> , pathognomonic. <b>Chronic</b> — <b>vegetative endocarditis of the LEFT AV (mitral) valve</b> and <b>proliferative non-suppurative arthritis</b> . Zoonotic (erysipeloid in fish handlers)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>BRUCELLOSIS</b> ★    | <i>Brucella abortus</i> (cattle), <i>B. melitensis</i> (goats, sheep — the most pathogenic for man), <i>B. suis</i> , <i>B. ovis</i> (rams). Facultative <b>intracellular</b> , survives in macrophages; has a predilection for the <b>placenta and fetal fluids because of ERYTHRITOL</b> . Lesions: <b>abortion in the LAST TRIMESTER</b> , retained placenta and metritis; <b>necrotic yellow leathery cotyledons with a THICKENED, OEDEMATOUS INTERCOTYLEDONARY MEMBRANE</b> covered in sticky exudate; fetal <b>bronchopneumonia</b> ; in bulls <b>orchitis and epididymitis with testicular necrosis</b> and <b>seminal vesiculitis</b> ; carpal/knee hygroma in Indian cattle. Tests: <b>RBPT (screening)</b> , <b>STAT</b> , <b>milk ring test</b> , <b>CFT</b> , <b>ELISA</b> ; culture of fetal <b>abomasal content</b> is the best sample. <b>Zoonosis</b> — <b>undulant/Malta fever</b> |

## 15 · CORYNEBACTERIUM · CAMPYLOBACTER · SALMONELLA · COLI · NECROBACILLOSIS

| Disease                      | Key point                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Caseous lymphadenitis</b> | <i>Corynebacterium pseudotuberculosis</i> , sheep and goats; abscess of superficial nodes and lung with <b>concentric laminated "ONION-RING" pus</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Bovine pyelonephritis</b> | <i>Corynebacterium renale</i> — <b>ascending</b> infection; pus in the renal pelvis with radiating pale streaks in the medulla; also <b>pizzle rot</b> of wethers                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>CAMPYLOBACTERIOSIS</b>    | <i>Campylobacter fetus</i> subsp. <i>venerealis</i> — <b>bovine genital campylobacteriosis</b> ; <b>venereal</b> , causes <b>early embryonic death</b> , <b>repeat breeding and infertility</b> rather than late abortion; mild endometritis. <i>C. jejuni</i> — enteritis and <b>ovine abortion</b> ; a major <b>food-borne zoonosis</b>                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>SALMONELLOSIS</b>         | <i>S. Dublin</i> and <i>S. Typhimurium</i> in calves; <i>S. Abortusequi</i> (equine abortion), <i>S. Abortusovis</i> . Lesions: <b>fibrinonecrotic ("diphtheritic") enterocolitis</b> with a yellow-grey adherent pseudomembrane, <b>"PARATYPHOID NODULES"</b> = pinpoint pale foci of necrosis in the <b>liver</b> , cholecystitis with a thickened gall bladder, splenomegaly, septicaemia in the young. <b>Zoonosis</b>                                                                                                                                                                                                                                                                                                                                                 |
| <b>COLIBACILLOSIS</b>        | <i>E. coli</i> . ① <b>Enterotoxigenic (ETEC)</b> — fimbriae <b>K99 (F5) in calves and lambs</b> , <b>K88 (F4) and F18 in piglets</b> ; enterotoxin causes <b>SECRETORY diarrhoea with a NORMAL villus</b> (no villous atrophy) → profuse watery white diarrhoea, dehydration, acidosis. ② <b>Septicaemic colibacillosis</b> — follows <b>failure of passive transfer</b> ; polyarthritis, meningitis, omphalitis. ③ <b>OEDEMA DISEASE of pigs</b> ★ — <b>Shiga toxin 2e</b> from F18 <i>E. coli</i> soon after <b>weaning</b> ; <b>oedema of the EYELIDS</b> , <b>forehead</b> , <b>submucosa of the GASTRIC CARDIA and MESOCOLON</b> , hydrothorax and ascites; staggering, squealing voice, sudden death; angiopathy with fibrinoid necrosis. ④ <b>Coliform mastitis</b> |
| <b>NECROBACILLOSIS</b>       | <i>Fusobacterium necrophorum</i> — the common denominator of <b>calf diphtheria</b> (necrotic laryngitis and oral necrosis), <b>foot rot</b> (with <i>Dichelobacter nodosus</i> ), <b>rumenitis-liver abscess complex</b> , and <b>necrotic rhinitis (bull nose) of pigs</b> . Foul odour and yellow-grey dry necrotic tissue                                                                                                                                                                                                                                                                                                                                                                                                                                              |

## 16 · MYCOPLASMA · CHLAMYDIA · Q-FEVER · ANAPLASMA · EHRLICHIA ★★

| Disease                                           | Agent & lesion                                                                                                                                                                                                                                                                                                                                                                                                                    |
|---------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>CBPP</b><br>Contagious bovine pleuropneumonia  | <i>Mycoplasma mycoides</i> subsp. <i>mycoides</i> (small colony type). Notifiable. Unilateral fibrinous pleuropneumonia: massive thickening of the INTERLOBULAR SEPTA with hepatisation of different ages gives the classic "MARBLED" cut surface; litres of straw-coloured fluid with fibrin clots in the pleural cavity; chronic cases form SEQUESTRA — encapsulated necrotic lung that keeps live organisms and makes carriers |
| <b>CCPP</b><br>Contagious caprine pleuropneumonia | <i>Mycoplasma capripneumoniae</i> in goats; acute fibrinous pleuropneumonia, usually unilateral, with pleural fluid; high mortality; no marbling and no sequestra as a rule                                                                                                                                                                                                                                                       |
| <b>Porcine enzootic pneumonia</b>                 | <i>Mycoplasma hyopneumoniae</i> . Cranioventral purple-red plum-coloured consolidation with sharp demarcation; the histological hallmark is "CUFFING PNEUMONIA" — heavy peribronchiolar and perivascular LYMPHOID hyperplasia. High morbidity, low mortality, poor growth                                                                                                                                                         |
| <b>Chlamydial diseases</b>                        | <i>Chlamydia abortus</i> → enzootic abortion of ewes, necrotic placentitis. <i>C. psittaci</i> → psittacosis/ornithosis in birds — fibrinous airsacculitis, pericarditis, hepatosplenomegaly; zoonotic. Also polyarthritis, conjunctivitis, sporadic bovine encephalomyelitis. Obligate intracellular; elementary and reticulate bodies, seen with Giemsa or Macchiavello stain                                                   |
| <b>Q FEVER</b>                                    | <i>Coxiella burnetii</i> — extremely resistant, spread by aerosol from birth fluids and by ticks; causes abortion and placentitis in ruminants (often subclinical) but serious influenza-like disease and endocarditis in people. A major occupational zoonosis of veterinarians                                                                                                                                                  |
| <b>ANAPLASMOSIS</b> ★                             | <i>Anaplasma marginale</i> in the red cell, tick-borne. Causes EXTRAVASCULAR haemolysis — therefore marked anaemia and JAUNDICE but NO haemoglobinuria. Pale thin blood, enlarged spleen, distended gall bladder, icteric carcass. Seen as a small dense dot at the MARGIN of the RBC                                                                                                                                             |
| <b>EHRLICHIOSIS</b>                               | <i>Ehrlichia canis</i> → tropical canine pancytopenia, spread by <i>Rhipicephalus sanguineus</i> . Thrombocytopenia with epistaxis and petechiae, anaemia, lymphadenopathy, splenomegaly; MORULAE in monocytes. Common in Indian dogs                                                                                                                                                                                             |

**HAEMOGLOBINURIA — YES OR NO? ★★★** BABESIOSIS — YES (intravascular haemolysis, red urine). ANAPLASMOSIS — NO (extravascular haemolysis, jaundice only). THEILERIOSIS — usually NO. Leptospirosis and post-parturient haemoglobinuria and chronic copper poisoning — YES. This one line settles many objective questions.

## 17 · SUPERFICIAL &amp; DEEP MYCOSES

| Mycosis                              | Agent & diagnostic feature                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|--------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>RINGWORM</b><br>(dermatophytosis) | <i>Trichophyton verrucosum</i> — the usual cause in CATTLE, thick grey-white crusty circular plaques around the eyes and neck. <i>Microsporum canis</i> — dogs and cats, and the one that shows apple-green FLUORESCENCE under WOOD'S (UV) LAMP. <i>T. mentagrophytes</i> , <i>M. gypseum</i> . Lesion: circular expanding alopecia with scaling and a healing centre; hyphae and arthrospores confined to the keratin and hair shaft, folliculitis. Diagnosis: 10–20% KOH mount of hair/scale, Wood's lamp, culture on DTM/Sabouraud. Zoonotic |
| <b>ASPERGILLOSIS</b>                 | <i>Aspergillus fumigatus</i> ; septate, dichotomously branching hyphae at about 45°. Bovine mycotic abortion ★ — thickened leathery necrotic placenta with raised plaques, and discrete raised "ringworm-like" plaques on the SKIN of the aborted fetus. Guttural pouch mycosis of horses — erodes the internal carotid → fatal epistaxis. Brooder pneumonia in chicks. Angioinvasive with infarction. Stain: PAS or GMS                                                                                                                        |
| <b>ZYGOMYCOSIS</b><br>(mucormycosis) | <i>Rhizopus</i> , <i>Mucor</i> , <i>Absidia</i> . Broad, irregular, NON-SEPTATE (sparsely septate) hyphae branching at wide angles. Strongly angioinvasive → thrombosis and infarction; gastric and abomasal ulcers, lymph node granulomas, mycotic placentitis                                                                                                                                                                                                                                                                                 |
| <b>HISTOPLASMOSIS</b>                | <i>Histoplasma capsulatum</i> — small intracellular yeasts inside macrophages; granulomatous disease of the intestine, liver, spleen and lung of dogs. <i>H. farciminosum</i> causes EPIZOOTIC LYMPHANGITIS of horses — corded lymphatics with ulcerating nodules (must be told apart from glanders)                                                                                                                                                                                                                                            |
| <b>CRYPTOCOCCOSIS</b>                | <i>Cryptococcus neoformans</i> , associated with pigeon droppings. Yeast with a huge mucoid CAPSULE that produces a gelatinous, "soap-bubble" appearance. Nasal granuloma of the CAT ("Roman nose"), meningitis. Capsule shown by INDIA INK negative staining and by MUCICARMINE (positive) — a favourite question                                                                                                                                                                                                                              |
| <b>RHINOSPORIDIOSIS</b> ★            | <i>Rhinosporidium seeberi</i> — very common in INDIA (cattle, buffalo, horses, dogs and man), linked to stagnant water and mud. Pink, polypoid, friable nasal mass studded with white pinhead specks, which are the SPORANGIA packed with endospores. Bleeds easily, recurs after removal                                                                                                                                                                                                                                                       |
| <b>CANDIDIASIS</b>                   | <i>Candida albicans</i> — "THRUSH", a white pseudomembrane on the tongue and oesophagus; crop mycosis (sour crop) in birds; follows prolonged antibiotic use and immunosuppression. Shows budding yeasts, pseudohyphae and hyphae together                                                                                                                                                                                                                                                                                                      |

## 18 • MYCOTOXICOSES ★★★

| Mycotoxycosis            | Fungus, toxin, target                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>AFLATOXICOSIS</b>     | <i>Aspergillus flavus</i> and <i>A. parasiticus</i> → aflatoxin B <sub>1</sub> , the most toxic. On <b>groundnut cake, maize and cottonseed</b> stored damp — the classic Indian feed problem. <b>Target = LIVER; it is both hepatotoxic and HEPATOCARCINOGENIC.</b> <b>Acute:</b> centrilobular necrosis and haemorrhage, icterus, bleeding (↓ clotting factors). <b>Chronic:</b> firm pale fibrotic liver with <b>marked BILE DUCT (biliary) HYPERPLASIA, MEGALOCYTOSIS</b> of hepatocytes, and eventually <b>hepatocellular carcinoma</b> . <b>Ducklings are the most sensitive species;</b> also immunosuppression and poor growth. Aflatoxin M <sub>1</sub> is excreted in <b>milk</b> |
| <b>OCHRATOXICOSIS</b>    | <i>Aspergillus ochraceus</i> , <i>Penicillium verrucosum</i> → <b>ochratoxin A</b> . <b>Target = KIDNEY (nephrotoxic).</b> Pale, enlarged, mottled kidneys with <b>tubular degeneration and interstitial fibrosis</b> — "mycotoxic porcine nephropathy"; polyuria, poor growth. Immunosuppressive                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>TRICHOTHECOSIS</b>    | <i>Fusarium</i> species → <b>T-2 toxin, deoxynivalenol (vomitoxin), diacetoxyscirpenol</b> . Trichothecenes are <b>protein-synthesis inhibitors and are RADIOMIMETIC: oral and gastrointestinal necrosis and ulceration</b> , feed refusal and vomiting, plus <b>necrosis of bone marrow and lymphoid tissue</b> → <b>pancytopenia and haemorrhage</b> . Also contact dermatitis of the muzzle                                                                                                                                                                                                                                                                                              |
| <b>DEGNALA DISEASE ★</b> | An <b>Indian condition of buffaloes and cattle in the northern states</b> , linked to feeding <b>mouldy PADDY (rice) STRAW</b> and its <i>Fusarium</i> toxins. Produces <b>DRY GANGRENE OF THE EXTREMITIES</b> — oedema then necrosis and sloughing of the <b>tail tip, ear tips, muzzle and coronary band/hoof</b> , with sharply demarcated dry black tissue. Not febrile; response is reported to <b>selenium plus vitamin E</b> and change of fodder                                                                                                                                                                                                                                    |
| <b>ERGOTISM</b>          | <i>Claviceps purpurea</i> on rye and other grasses → <b>ergot alkaloids (ergotamine, ergotoxine)</b> which are powerful <b>VASOCONSTRICTORS</b> → <b>DRY GANGRENE</b> of the <b>ear tips, tail, teats and feet</b> , agalactia, abortion; a <b>convulsive/nervous form</b> also occurs. <i>Claviceps fusiformis</i> on <b>BAJRA (pearl millet)</b> in India causes <b>AGALACTIA</b> in sows and buffaloes                                                                                                                                                                                                                                                                                   |

**ONE-LINE TARGETS** Aflatoxin → **LIVER**. Ochratoxin → **KIDNEY**. Trichothecene → **GUT + BONE MARROW**. Fumonisin → **BRAIN** of the horse (leukoencephalomalacia). Zearalenone → **OESTROGENIC** (vulvar oedema, prolapse in gilts). Ergot → **BLOOD VESSELS** (gangrene).

## 19 • HAEMOPROTOZOA ★★★ (HIGHEST YIELD IN INDIA)

|                | BABESIOSIS                                                                                                                                                                                                                                                              | THEILERIOSIS                                                                                                                                                                                                                                                 | TRYPANOSOMOSIS                                                                                                                                                                                                                                 |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Agent          | <i>Babesia bigemina</i> , <i>B. bovis</i> ; <i>B. canis</i> in dogs                                                                                                                                                                                                     | <i>Theileria annulata</i> (tropical theileriosis of India), <i>T. parva</i> (East Coast fever)                                                                                                                                                               | <i>Trypanosoma evansi</i> = <b>SURRA</b> ; also <i>T. congolense</i> , <i>T. vivax</i>                                                                                                                                                         |
| Vector         | <i>Rhipicephalus (Boophilus) microplus</i> tick; <b>transovarial</b>                                                                                                                                                                                                    | <i>Hyalomma</i> tick; <b>transstadial</b>                                                                                                                                                                                                                    | <b>Biting flies (Tabanus)</b> <b>MECHANICALLY</b> ; also iatrogenic needles                                                                                                                                                                    |
| Where it lives | <b>Inside the RBC</b> — large <b>PEAR-SHAPED PAIRS</b> at an acute angle                                                                                                                                                                                                | <b>Schizonts (KOCH'S BLUE BODIES) in LYMPHOCYTES</b> , then small piroplasms in RBCs                                                                                                                                                                         | <b>EXTRACELLULAR</b> flagellate in plasma                                                                                                                                                                                                      |
| Key signs      | High fever, <b>intravascular haemolysis</b> → <b>HAEMOGLOBINURIA ("redwater")</b> , jaundice, anaemia. <i>B. bovis</i> also causes <b>CEREBRAL babesiosis</b> — parasitised cells sequester in brain capillaries, giving a <b>"cherry-pink" brain</b> and nervous signs | <b>Marked enlargement of superficial lymph nodes first</b> (parotid node), fever, anaemia, jaundice, <b>petechiae, "PUNCHED-OUT" ABOMASAL ULCERS</b> , pulmonary oedema, <b>lymphoid proliferation then destruction</b> . Exotic breeds and crossbreds worst | Intermittent fever, progressive anaemia and <b>cachexia</b> , ventral oedema, <b>corneal opacity in dogs</b> , splenomegaly; nervous signs late. <b>Horses, camels and dogs are highly susceptible</b> ; cattle and buffalo are often carriers |
| Diagnosis      | <b>Giemsa-stained peripheral blood smear from the EAR-TIP capillary</b>                                                                                                                                                                                                 | <b>Lymph node biopsy/aspirate smear for Koch's blue bodies + blood smear</b>                                                                                                                                                                                 | <b>Wet blood film for motile organisms</b> , Giemsa smear, <b>mouse inoculation</b> , CATT, PCR                                                                                                                                                |

## 20 • FLUKES, WORMS, TAPEWORMS, COCCIDIA

| Parasite              | Pathology                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>FASCIOLOSIS ★</b>  | <i>Fasciola hepatica</i> and <i>F. gigantica</i> (the important one in India); snail host <i>Lymnaea</i> .<br><b>Acute:</b> immature flukes tunnel through the parenchyma → <b>haemorrhagic migratory tracts</b> , enlarged friable liver, fatal haemorrhage; also creates the anaerobic focus for <b>BLACK DISEASE</b> .<br><b>Chronic:</b> adults in the bile ducts → <b>cholangitis</b> , marked <b>biliary fibrosis and calcification</b> = "PIPE-STEM LIVER", cirrhosis, <b>black parasitic haematin pigment</b> , <b>anaemia</b> , <b>hypoalbuminaemia</b> and <b>BOTTLE JAW</b> |
| <b>AMPHISTOMIASIS</b> | <i>Paramphistomum</i> . The <b>IMMATURE flukes are the killers</b> — they attach in the <b>duodenum</b> causing severe haemorrhagic <b>duodenitis</b> , oedema and death in young stock; the <b>ADULTS in the rumen are relatively harmless</b> (rumen papillae studded with pink flukes)                                                                                                                                                                                                                                                                                              |
| <b>Ascariasis</b>     | <i>Ascaris suum</i> in pigs — larval migration leaves " <b>MILK SPOTS</b> " (white fibrotic foci) on the <b>LIVER</b> , a condemnation lesion; larvae also cause verminous pneumonia. <i>Parascaris equorum</i> in foals — intestinal obstruction and rupture. <i>Toxocara canis</i> — zoonotic <b>visceral larva migrans</b>                                                                                                                                                                                                                                                          |
| <b>Strongylosis</b>   | <i>Strongylus vulgaris</i> — larvae migrate in the <b>CRANIAL MESENTERIC ARTERY</b> → <b>verminous arteritis and aneurysm</b> , thrombosis, <b>intestinal infarction and fatal colic</b> in horses                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>HAEMONCHOSIS</b>   | <i>Haemonchus contortus</i> , the " <b>BARBER'S POLE WORM</b> " of the <b>abomasum</b> . A <b>voracious blood-sucker</b> → severe <b>anaemia</b> , <b>hypoproteinaemia</b> and <b>BOTTLE JAW</b> , usually <b>WITHOUT diarrhoea</b> ; pale watery carcass. Abomasal mucosa studded with worms and small bleeding points                                                                                                                                                                                                                                                                |
| <b>SPIROCERCOSIS</b>  | <i>Spirocerca lupi</i> in the <b>dog's OESOPHAGUS</b> (beetle intermediate host) → firm nodules with a central pore that may transform into <b>FIBROSARCOMA</b> or <b>OSTEOSARCOMA</b> ; also <b>aortic scarring/aneurysm</b> and <b>spondylitis of the caudal thoracic vertebrae</b>                                                                                                                                                                                                                                                                                                  |

|                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Filariasis</b>        | <i>Setaria</i> — harmless in the peritoneum but <b>aberrant migration into the spinal cord causes CEREBROSPINAL NEMATODIASIS ("KUMRI") in goats and horses in India</b> — sudden hind-limb weakness. <i>Dirofilaria immitis</i> — heartworm in the <b>right ventricle and pulmonary artery</b> , pulmonary endarteritis, <b>caval syndrome</b> . <i>Parafilaria bovicola</i> — "summer bleeding" nodules on cattle skin. <i>Onchocerca</i> — ligament and skin                                                                                                                  |
| <b>Hookworm</b>          | <i>Ancylostoma caninum</i> — severe <b>iron-deficiency anaemia and melena</b> in pups; percutaneous, transmammary and oral routes                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Tapeworms</b>         | <i>Cysticercus bovis</i> ( <i>Taenia saginata</i> ) in beef <b>muscle, heart and tongue</b> — "measly beef". <i>Cysticercus cellulosae</i> ( <i>T. solium</i> ) in pork — and <b>human neurocysticercosis</b> . <i>Echinococcus granulosus</i> → <b>HYDATID CYST</b> in the liver and lung of herbivores and man; the <b>dog is the definitive host</b> ; the cyst has a laminated membrane and brood capsules with <b>hydatid sand</b> . <i>Coenurus cerebralis</i> ( <i>T. multiceps</i> ) → " <b>GID</b> "/sturdy in sheep brain. <i>Moniezia</i> in ruminants, usually mild |
| <b>Coccidiosis</b>       | <i>Eimeria zuernii</i> and <i>E. bovis</i> in calves — <b>haemorrhagic typhlocolitis</b> , bloody diarrhoea with <b>tenesmus and rectal prolapse</b> ; the parasite destroys the epithelium of the caecum and colon                                                                                                                                                                                                                                                                                                                                                             |
| <b>Toxoplasmosis</b>     | <i>Toxoplasma gondii</i> ; the <b>CAT is the only definitive host</b> (sheds oocysts). Causes <b>abortion in SHEEP with small WHITE NECROTIC FOCI scattered over the COTYLEDONS</b> ("strawberry jam" placenta), fetal brain and liver necrosis; encephalitis and pneumonia in other species. <b>Zoonosis</b> — dangerous in pregnancy                                                                                                                                                                                                                                          |
| <b>Cryptosporidiosis</b> | <i>Cryptosporidium parvum</i> — a major cause of <b>calf diarrhoea at 1–3 weeks</b> ; attaches to the microvillous border of the ileum → <b>villous atrophy and malabsorption</b> . Oocysts are <b>acid-fast</b> (modified Ziehl–Neelsen); <b>water-borne ZOONOSIS</b> , resistant to chlorine                                                                                                                                                                                                                                                                                  |

## 21 · NUTRITIONAL IMBALANCES ★★

**Energy & macronutrients** — **Carbohydrate excess** → **ruminal lactic acidosis** (grain overload) with rumenitis and liver abscess; **carbohydrate deficit** → **ketosis** and pregnancy toxemia. **Protein deficiency** → hypoproteinaemic **oedema and ascites**, poor growth, immunosuppression. **Fat** — essential fatty acid deficiency gives scaly skin; **steatitis ("yellow fat disease")** occurs in cats fed oily fish with **vitamin E deficiency**.

| Mineral                         | Deficiency disease                                                                                                                                                                                                                                                                                                                            |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Ca / P</b>                   | <b>Rickets</b> (young), <b>osteomalacia</b> (adult), <b>fibrous osteodystrophy</b> when Ca:P is wrong, <b>milk fever</b> (parturient hypocalcaemia)                                                                                                                                                                                           |
| <b>Magnesium</b>                | <b>HYPOMAGNESAEMIC TETANY</b> ("grass tetany/staggers") — lush spring grass, lactating cows; tetany, convulsions, sudden death                                                                                                                                                                                                                |
| <b>COPPER</b>                   | <b>SWAY BACK / enzootic ataxia</b> in lambs (cerebral white-matter cavitation, spinal demyelination) · <b>achromotrichia</b> — loss of coat colour with <b>"SPECTACLE EYE"</b> in cattle · anaemia · <b>"falling disease"</b> from myocardial fibrosis · poor wool crimp ("steely wool"). Made worse by <b>excess molybdenum</b> and sulphate |
| <b>Cobalt</b>                   | Needed for vitamin B <sub>12</sub> synthesis in the rumen → <b>"pine"/wasting</b> , and <b>WHITE LIVER DISEASE</b> of lambs (fatty liver)                                                                                                                                                                                                     |
| <b>Iodine</b>                   | <b>GOITRE</b> — enlarged thyroid; newborns <b>weak, HAIRLESS, stillborn with a swollen neck</b> ; goitrogens in brassicas and soybean                                                                                                                                                                                                         |
| <b>SELENIUM (+ vit E)</b>       | <b>WHITE MUSCLE DISEASE, hepatosis dietetica</b> of pigs, <b>MULBERRY HEART DISEASE</b> of pigs, exudative diathesis of chicks, retained placenta, poor immunity                                                                                                                                                                              |
| <b>Zinc</b>                     | <b>PARAKERATOSIS</b> of pigs — symmetrical crusty scaly skin; thymic atrophy, poor wound healing                                                                                                                                                                                                                                              |
| <b>Iron</b>                     | <b>Baby-pig anaemia</b> — microcytic hypochromic; the classic reason for iron injection at birth                                                                                                                                                                                                                                              |
| Vitamin                         | Deficiency lesion                                                                                                                                                                                                                                                                                                                             |
| <b>A</b>                        | <b>Night blindness</b> → <b>xerophthalmia, SQUAMOUS METAPLASIA and keratinisation</b> of epithelium, <b>hydrocephalus and blindness from stenosis of the optic canal in calves</b> , pituitary cysts, poor immunity, papilloedema                                                                                                             |
| <b>D</b>                        | <b>Rickets and osteomalacia</b>                                                                                                                                                                                                                                                                                                               |
| <b>E</b>                        | See selenium; also <b>steatitis</b> and reproductive failure                                                                                                                                                                                                                                                                                  |
| <b>K</b>                        | <b>Bleeding</b> — sweet clover (dicoumarol), warfarin, prolonged sulphonamide use                                                                                                                                                                                                                                                             |
| <b>B<sub>1</sub> thiamine</b>   | <b>POLIOENCEPHALOMALACIA</b> in ruminants (thiaminase, high grain, amprolium, bracken fern in horses); <b>"Chastek paralysis"</b> in fur animals and cats fed <b>raw fish thiaminase</b>                                                                                                                                                      |
| <b>B<sub>2</sub> riboflavin</b> | <b>"CURLED-TOE PARALYSIS"</b> of chicks — peripheral nerve demyelination                                                                                                                                                                                                                                                                      |
| <b>Niacin</b>                   | <b>"BLACK TONGUE"</b> of dogs, pellagra-like; maize-based diets                                                                                                                                                                                                                                                                               |
| <b>Choline / Manganese</b>      | <b>PEROSIS ("slipped tendon")</b> in chicks                                                                                                                                                                                                                                                                                                   |
| <b>C</b>                        | <b>SCURVY</b> — only in <b>GUINEA PIGS, primates and fish</b> , which cannot synthesise it; haemorrhage, poor collagen, loose teeth                                                                                                                                                                                                           |

## 22 · METABOLIC DISEASES ★★

| Disease                                  | Mechanism & lesion                                                                                                                                                                                                                                                                                                                                                                        |
|------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>PREGNANCY TOXAEMIA</b>                | <b>Ewes and does in LATE gestation carrying twins/triplets</b> ; negative energy balance → fat mobilisation → <b>ketosis and severe FATTY LIVER</b> (huge, pale yellow, greasy, friable). Ketonuria, hypoglycaemia, blindness, recumbency. Also seen in fat queens                                                                                                                        |
| <b>Ketosis of dairy cattle</b>           | Early lactation, high yield; ketone smell on breath and milk, fatty liver, low milk yield; <b>Rothera's test on urine/milk</b>                                                                                                                                                                                                                                                            |
| <b>POST-PARTURIENT HAEMOGLOBINURIA ★</b> | <b>HYPOPHOSPHATAEMIA</b> in high-yielding <b>buffaloes and cows within a few weeks of calving</b> , often on rice straw/berseem with low phosphorus. <b>Intravascular haemolysis</b> → <b>haemoglobinuria, severe anaemia, pale mucosae, jaundice, pale fatty liver</b> . Responds to <b>phosphorus</b> therapy. Distinguish from babesiosis (which has parasites on the smear and fever) |
| <b>Hypomagnesaemic tetany</b>            | As above; <b>often no gross lesion</b> — diagnosis from serum/CSF magnesium and history                                                                                                                                                                                                                                                                                                   |
| <b>AZOTURIA</b>                          | <b>Exertional rhabdomyolysis</b> of horses: full grain ration during rest, then sudden work. <b>Hard, painful, swollen gluteal muscles</b> , sweating, reluctance to move, <b>coffee-coloured MYOGLOBINURIC urine</b> ; pale streaked muscle at necropsy; ↑ CK and AST. <b>Plasma remains clear</b>                                                                                       |
| <b>SWAY BACK / enzootic ataxia</b>       | <b>Copper</b> deficiency in lambs — congenital form present at birth, delayed form later; white-matter cavitation and demyelination                                                                                                                                                                                                                                                       |
| <b>PICA</b>                              | Depraved appetite from <b>phosphorus and mineral deficiency</b> ; chewing bones and soil → leads to <b>BOTULISM</b> and foreign-body/hardware disease                                                                                                                                                                                                                                     |
| <b>Rheumatism-like syndrome</b>          | Stiffness, shifting lameness and joint pain without a single defined agent; polyarthritis, periarticular fibrosis                                                                                                                                                                                                                                                                         |

## 23 • TOXICITIES ★★★

| Poison                     | The lesion / clue that names it                                                                                                                                                                                                                                                                                                                                                              |
|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ARSENIC</b>             | Inorganic: severe <b>haemorrhagic gastroenteritis</b> , profuse <b>"RICE-WATER" diarrhoea</b> , <b>bright red velvety abomasal/gastric mucosa</b> , garlicky odour of the ingesta, fatty liver and tubular nephrosis. Organic phenylarsonic feed additives in pigs → <b>peripheral nerve demyelination, blindness and ataxia</b>                                                             |
| <b>COPPER (chronic)</b>    | <b>SHEEP</b> are highly susceptible; copper accumulates in the liver, then a stressor triggers a sudden <b>HAEMOLYTIC CRISIS</b> — jaundice, haemoglobinuria, <b>enlarged friable yellow-orange liver</b> and dark <b>"GUN-METAL" KIDNEYS</b> . Confirm by liver and kidney copper assay                                                                                                     |
| <b>LEAD</b>                | <b>Commonest in cattle and calves</b> — discarded batteries, paint, used engine oil. <b>Nervous disease: blindness, head-pressing, bellowing, convulsions, champing of jaws</b> ; also anaemia. <b>Blood smear: BASOPHILIC STIPPLING with nucleated red cells. Acid-fast intranuclear inclusions in renal tubular epithelium.</b> Confirm on blood, liver and kidney lead                    |
| <b>MERCURY</b>             | <b>Organic</b> (mercury-treated <b>seed grain</b> ) → neurological — ataxia, blindness, laminar cortical necrosis, peripheral neuropathy. <b>Inorganic</b> → corrosive gastroenteritis + <b>severe renal tubular necrosis</b>                                                                                                                                                                |
| <b>CADMIUM</b>             | <b>Nephrotoxic</b> (proximal tubule), <b>testicular degeneration and necrosis</b> , anaemia, poor growth, interferes with zinc and copper                                                                                                                                                                                                                                                    |
| <b>STRYCHNINE</b>          | <b>Blocks the inhibitory transmitter GLYCINE in the spinal cord</b> → violent <b>TETANIC convulsions with extensor rigidity, "sawhorse" stance and risus sardonicus, TRIGGERED BY NOISE, LIGHT OR TOUCH</b> ; the animal stays <b>conscious</b> ; death from respiratory paralysis; <b>rigor mortis sets in very quickly</b> . Essentially <b>no gross lesions</b> — analyse stomach content |
| <b>NITRATE / NITRITE ★</b> | Nitrate in oat hay, sorghum, fertiliser and water is <b>reduced to NITRITE in the rumen</b> ; nitrite converts haemoglobin to <b>METHAEMOGLOBIN</b> , which cannot carry oxygen. <b>Blood and tissues are CHOCOLATE-BROWN</b> , mucosae are muddy-                                                                                                                                           |

|                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>HYDROCYANIC ACID ★</b>          | brown/cyanotic, dyspnoea, rapid death; also abortion. <b>Antidote: METHYLENE BLUE</b> (and ascorbic acid)<br><b>Cyanogenic glycosides in young, wilted, drought-stressed or frosted SORGHUM/Johnson grass and cassava.</b> Cyanide blocks <b>cytochrome oxidase</b> , so tissues cannot use oxygen. <b>Blood stays BRIGHT CHERRY-RED</b> , mucosae bright red, <b>bitter-almond odour</b> of the rumen contents, death in minutes. <b>Antidote: sodium nitrite followed by sodium thiosulphate</b> |
| <b>FLUORIDE</b>                    | <b>Dental fluorosis</b> — <b>mottled, chalky, pitted, brown-stained, rapidly worn teeth, but ONLY those teeth that were developing during exposure; exostoses (bony outgrowths) on the mandible and metatarsals</b> , stiffness and lameness. Sources: industrial fallout and high-fluoride bore-well water — endemic in parts of India                                                                                                                                                            |
| <b>SELENIUM (excess)</b>           | <b>"ALKALI DISEASE"</b> (chronic) — loss of hair from the <b>mane and tail</b> , and <b>deformed, sloughing hooves with a circular break at the coronet. "BLIND STAGGERS"</b> (subacute) — wandering, blindness, paralysis. From selenium-accumulating plants on seleniferous soil (Punjab/Haryana belt)                                                                                                                                                                                           |
| <b>OXALATES</b>                    | In <b>Oxalis, Setaria, Halogeton, amaranthus, sugar beet tops</b> . Bind calcium → <b>acute hypocalcaemia and tetany</b> , and <b>calcium oxalate crystals in the renal tubules</b> → <b>NEPHROSIS and renal failure</b> . Chronic low intake with grass leads to bone changes                                                                                                                                                                                                                     |
| <b>ORGANOPHOSPHATE / CARBAMATE</b> | <b>Inhibit acetylcholinesterase</b> → acetylcholine accumulates. <b>Muscarinic "SLUD"</b> — Salivation, Lacrimation, Urination, Defecation, plus <b>miosis, bradycardia, bronchoconstriction and pulmonary oedema</b> ; nicotinic — muscle tremor then paralysis; CNS — convulsions. Diagnosis: <b>low blood cholinesterase. Antidote: ATROPINE; add 2-PAM (pralidoxime) for organophosphates but NOT for carbamates</b>                                                                           |
| <b>ORGANOCHLORINE</b>              | DDT, BHC/lindane, endosulfan — <b>CNS STIMULATION</b> : hyperexcitability, muscle tremors beginning at the head, exaggerated response to stimuli, convulsions. Highly <b>lipophilic and persistent</b> — <b>residues in fat and milk</b> . No specific antidote                                                                                                                                                                                                                                    |

## 24 · PLANT POISONING ★★★

| Plant                                             | Toxin & pathology                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|---------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>BRACKEN FERN</b><br><i>Pteridium aquilinum</i> | Toxin <b>PTAQUILOSIDE</b> (radiomimetic) plus <b>THIAMINASE</b> . Three quite different pictures — know all three:<br><b>CATTLE (acute)</b> — bone marrow <b>APLASIA</b> → <b>pancytopenia, thrombocytopenia and a haemorrhagic syndrome</b> with bleeding from all orifices and secondary infection.<br><b>CATTLE (chronic)</b> → <b>ENZOOTIC HAEMATURIA</b> — bladder <b>haemangiomas and carcinomas</b> with persistent haematuria (bovine papillomavirus acts as a co-factor).<br><b>HORSES</b> — <b>thiaminase</b> → <b>thiamine deficiency</b> , ataxia, "bracken staggers".<br><b>SHEEP</b> — <b>"BRIGHT BLINDNESS"</b> from progressive retinal degeneration |
| <b>GOSSYPOL</b><br>(cottonseed cake)              | A polyphenol in <b>cottonseed</b> ; toxic to <b>monogastrics and young ruminants</b> (adult rumen binds it). <b>Cardiotoxic and hepatotoxic</b> → myocardial degeneration and heart failure, giving <b>generalised oedema: pulmonary oedema, hydrothorax, hydropericardium and ascites</b> ; centrilobular hepatic necrosis; anaemia; <b>reduced fertility in males</b> ; in layers, <b>olive-green/discoloured yolks and mottled eggs</b>                                                                                                                                                                                                                           |
| <b>RATTI</b><br><i>Abrus precatorius</i>          | The red-and-black "crab's-eye" seed. Toxin <b>ABRIN</b> , a <b>toxalbumin / ribosome-inactivating protein</b> (like ricin). In India the crushed seed paste is made into a <b>"SUI" (needle) and driven under the skin to poison cattle deliberately</b> — a <b>veterolegal</b> problem. Produces <b>local oedema and necrosis at the site</b> , then <b>severe haemorrhagic gastroenteritis, agglutination and haemolysis of red cells, hepatic and renal degeneration</b> and death in 1–3 days. At necropsy search the skin for the puncture and recover the needle                                                                                               |
| <b>LANTANA</b><br><i>Lantana camara</i>           | Toxin <b>LANTADENE A (and B)</b> — the classic Indian hepatotoxin of cattle, buffalo and sheep browsing the weed. Causes <b>intrahepatic CHOLESTASIS and hepatocellular necrosis</b> → <b>jaundice, ruminal stasis and severe constipation</b> , and above all <b>HEPATOGENOUS (Type III) PHOTSENSITIZATION</b> because <b>phylloerythrin</b> is no longer excreted: crusting, peeling and necrosis of <b>the muzzle ("PINK NOSE"), eyelids, ears and any unpigmented skin</b> . Also renal tubular damage. Liver is enlarged, yellow-orange and firm; gall bladder distended                                                                                        |

## 25 · BLANK-FILL DRILL — UNIT 4

- FMD virus is an Aphthovirus of family Picornaviridae with 7 serotypes
- Serotypes of FMD circulating in India are O, A and Asia-1
- Tiger heart in calves occurs in FMD
- Of the vesicular diseases, only vesicular stomatitis affects horses
- Zebra striping of the colon is seen in rinderpest and PPR
- The microscopic hallmark of malignant catarrhal fever is necrotising lymphocytic vasculitis
- Bluetongue is spread by Culicoides and shows haemorrhage at the base of the pulmonary artery
- IBR virus becomes latent in the trigeminal ganglion
- Persistently infected BVD calves arise from non-cytopathic infection at 40–120 days of gestation
- Linear oesophageal erosions suggest mucosal disease
- EIA is diagnosed by the Coggins test and shows sideroleukocytes
- EHV-1 abortion produces a fresh (non-autolysed) fetus with hepatic intranuclear inclusions
- Button ulcers, marginal splenic infarcts and turkey-egg kidney = classical swine fever
- Rabies virus travels to the CNS centripetally along peripheral nerves
- The gold-standard test for rabies is the fluorescent antibody test on brain
- Hard pad and enamel hypoplasia occur in canine distemper
- Gall-bladder wall oedema with Cowdry type A inclusions = infectious canine hepatitis
- Canine parvovirus destroys the intestinal crypts; rotavirus destroys the villus tips
- The wheelbarrow test is used for jaagsiekte (ovine pulmonary adenocarcinoma)
- Prions consist of protein only and cause status spongiosus with no inflammation
- Bollinger bodies occur in fowl pox; Guarnieri bodies in vaccinia
- Orf is caused by a parapoxvirus and is zoonotic
- Pearl disease refers to serosal tuberculosis in cattle
- Johne's disease shows no caseation and no ulceration, unlike tuberculosis
- Blackleg is endogenous and needs no wound; malignant oedema is exogenous
- Black disease requires liver damage by immature Fasciola
- Pulpy kidney of lambs is caused by Clostridium perfringens type D and its epsilon toxin
- Botulism causes flaccid paralysis; tetanus causes spastic paralysis
- Tetanospasmin blocks the release of glycine and GABA
- Star-shaped nasal scars and farcy buds occur in glanders; test = mallein
- Haemorrhagic septicaemia is caused by Pasteurella multocida serotypes B:2 and E:2
- Shipping fever is caused by Mannheimia haemolytica
- Diamond-shaped skin lesions = swine erysipelas
- Brucella localises in the placenta because of erythritol; screening test = RBPT
- Circling disease from brain-stem microabscesses = listeriosis, from feeding silage
- Paratyphoid nodules in the liver indicate salmonellosis
- Oedema disease of pigs is caused by Shiga toxin 2e and affects the eyelids and gastric cardia
- Marbled lung with sequestra = CBPP; cuffling pneumonia = Mycoplasma hyopneumoniae
- Anaplasmosis gives anaemia and jaundice but no haemoglobinuria
- Morulae in monocytes indicate Ehrlichia canis
- Microsporum canis fluoresces under the Wood's lamp

- Aspergillus hyphae are septate; zygomycete hyphae are non-septate
- Cryptococcus capsule is shown by India ink and stains with mucicarmine
- Nasal polyps with sporangia = rhinosporidiosis
- Aflatoxin targets the liver; ochratoxin the kidney; trichothecenes the gut and bone marrow
- Chronic aflatoxicosis causes bile duct hyperplasia and megalocytosis
- Degnala disease follows feeding mouldy paddy straw and causes dry gangrene of the extremities
- Ergot alkaloids cause gangrene by vasoconstriction; Claviceps fusiformis on bajra causes agalactia
- Babesia appears as pear-shaped pairs; Theileria schizonts are Koch's blue bodies
- Theileria annulata is transmitted by Hyalomma ticks and causes punched-out abomasal ulcers
- Surra is caused by Trypanosoma evansi spread mechanically by Tabanus flies
- Pipe-stem liver is chronic fasciolosis; the snail host is Lymnaea
- In amphistomiasis the immature flukes in the duodenum cause the fatal disease
- Milk spots on the pig liver are caused by Ascaris suum larvae
- Haemonchus contortus is the barber's pole worm of the abomasum
- Spirocerca lupi nodules may become fibrosarcoma or osteosarcoma
- Kumri in goats and horses is caused by aberrant Setaria larvae
- The definitive host of Echinococcus granulosus is the dog; the cyst contains hydatid sand
- Coenurus cerebralis causes gid in sheep
- Toxoplasma abortion in sheep shows white necrotic foci on the cotyledons; definitive host = cat
- Cryptosporidium oocysts are acid-fast and resistant to chlorine
- Curled-toe paralysis in chicks = riboflavin deficiency; perosis = manganese/choline
- Black tongue in dogs = niacin deficiency
- Scurvy occurs only in guinea pigs, primates and fish
- Spectacle eye in cattle indicates copper deficiency
- White liver disease of lambs = cobalt deficiency
- Post-parturient haemoglobinuria is caused by hypophosphataemia
- Rice-water diarrhoea with a red velvety abomasum = arsenic poisoning
- Gun-metal kidneys and a yellow friable liver in sheep = chronic copper poisoning
- Basophilic stippling with nucleated RBCs in a blind, head-pressing calf = lead poisoning
- Strychnine blocks glycine; convulsions are triggered by noise, light or touch
- Nitrite poisoning gives chocolate-brown blood; antidote methylene blue
- Cyanide poisoning gives bright cherry-red blood; antidote sodium nitrite + sodium thiosulphate
- Alkali disease is chronic selenium poisoning with loss of mane and tail hair
- Organophosphates inhibit acetylcholinesterase; antidote atropine plus 2-PAM
- Bracken fern toxin is ptaquiloside; it causes bright blindness in sheep
- Gossypol comes from cottonseed and is cardiotoxic
- Ratti seed toxin is abrin, used in the veterolegal sui (needle) method
- Lantana toxin is lantadene A and causes hepatogenous photosensitization